

Question # 1

Durank Co is a listed refrigerator manufacturing company based in Madonk, a central European country whose currency is the Madonk dollar (\$).

It is considering expanding by setting up a new factory in the neighbouring country of Flontland, where the currency is the Flontland florin (FF).

Durank Co's capital structure comprises 200 million shares trading at \$3 each, and \$100 million nominal value bonds trading at \$110 per \$100. It has an 'A' rating according to the biggest credit rating agency in Madonk, and this corresponds to a credit spread of 55 basis points. This is expected to stay unchanged as the company expands into Flontland.

Durank Co has never operated in Flontland before. However, one of its competitors, Kloonish Co, expanded into Flontland a few years ago and now operates successfully in both countries. The directors consider that the systematic risk facing Durank Co as it expands will be comparable to that facing Kloonish Co.

Kloonish Co has an equity beta of 1.57, and the debt beta is assumed to be approximately zero. It has 100 million shares in issue, each trading at \$2.40. It also has \$75 million of bonds in its statement of financial position, which are trading at \$106 per \$100 nominal value.

Assume that both Kloonish Co and Durank Co pay tax at an average rate of 30%. The risk free rate of interest is 3% and the equity risk premium is 5%.

The finance director has prepared the following preliminary forecasts for the new project in Flontland (all cash flows are forecast in nominal terms):

		Year 1	Year 2	Year 3	Year 4
Units to be sold (million)		10	11	13	12

FF millions	Now	Year 1	Year 2	Year 3	Year 4
Revenue		230	250	285	260
Relevant costs		180	199	224	202
Investment	200				
Residual value					100
Tax allowable depreciation (including final year balancing allowance)		10	10	10	70

	Now	Year 1	Year 2	Year 3	Year 4
Spot exchange rate (FF... to \$1)	4.25				
Flontland inflation		5%	4%	3%	2%
Madonk inflation		6%	6%	4%	4%

The only cash flows not included in the above analysis are:

Component cost

The new factory business unit in Flontland will need to purchase a component from Durank Co to incorporate within the refrigerator manufacturing process. Durank Co has agreed to fix the price of each component at \$0.50 per each unit produced and sold. The cost of this component should be treated as a tax-deductible expense in Flontland, and the contribution generated in Madonk (being 40% of revenue earned) should be treated as taxable income.

Extra tax due in Madonk

The tax rate in Flontland is 10%, compared to 30% in Madonk. Both countries' taxes are payable in the year that the tax liability arises, and any tax losses can be relieved in the same year. A bi-lateral tax treaty exists between the two countries. All net cash flows arising from the project will be remitted back to Durank Co at the end of each year.

The Durank Co directors have closely monitored the investment Kloonish Co made in Flontland a few years ago, and watched how the initial investment was supplemented by follow-on projects. They consider that if Durank Co's initial investment project is successful, a further investment of \$10 million in nominal terms after four years could enable the company to generate the following post-tax free cash flows (again forecast in nominal terms):

	Year 5	Year 6	Year 7	Year 8
Post-tax FCF (\$ million)	7	9	5	3

The volatility (standard deviation) associated with these cash flows has been estimated to be 20%.

(a) Discuss how real options theory can be used alongside net present value to help companies with their investment decisions.

(6 marks)

(b) Prepare a report for the board of directors of Durank Co which:

(i) calculates a suitable weighted average cost of capital that Durank Co can use to discount the Flontland project cash flows.

(7 marks)

(ii) estimates the net present value of the Flontland project in \$, ignoring the value of the follow-on project.

(11 marks)

(iii) estimates the value of the follow-on project in \$, using the Black Scholes Option Pricing model.

(4 marks)

(iv) discusses the assumptions made in the above calculations, and any further factors that would need to be considered before making a final decision on whether to go ahead with the Flontland investment.

(12 marks)

Professional marks will be awarded for the demonstration of skill in communication, analysis and evaluation, scepticism and commercial acumen in your answer.

(10 marks)

Question # 2

Amison Co is a listed wholesale company in the grocery sector.

The sector is becoming increasingly competitive as global food prices rise, but customers continue to demand low prices.

It has decided to vertically integrate into the grocery-retailing sector.

The directors have identified a small chain of supermarkets, Birks Co, that could be acquired. Initial conversations between the two boards of directors have been encouraging. The directors of Birks Co think that their shareholders will like the culture and ethos of Amison Co, and would be open to the possibility of a takeover. In recent years, there have been several mergers and acquisitions in the grocery sector, and premiums to target company shareholders have averaged approximately 20%.

Two alternative offers are being considered by the Amison Co directors:

- 1 A cash offer of \$2.40 per share.
- 2 A share for share exchange, offering 4 shares in the combined company in exchange for every 5 shares in Birks Co.

Amison Co has 100 million shares in issue, trading at \$3.35 per share. Its earnings were \$27.92 million in the most recent financial year.

Birks Co has 60 million shares in issue, and its earnings were \$11.61 million in the most recent financial year. As it is an unlisted company, it should be assumed that a suitable P/E ratio for Birks Co is 10% lower than Amison Co's P/E ratio.

If the acquisition is successful, the directors expect that post tax synergies of \$2.98 million per year will be achieved, and that the combined company's P/E ratio will be the same as Amison Co's P/E ratio before the acquisition.

(a) Discuss possible sources of the synergy arising from Arnison Co's acquisition of Birks Co.

(4 marks)

(b) Estimate the percentage gains to the Arnison Co and the Birks Co shareholders under each of the two offers. Comment on the answers obtained, and the position of each company's shareholders.

Note: Up to 5 marks are available for discussion.

(16 marks)

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, scepticism and commercial acumen in your answer.

(5 marks)

Question # 3

Lamontagne Co is a large multinational company.

It has grown rapidly over the last few years and now has subsidiaries in several different countries.

Today is 1 June. The finance manager of the South Asian subsidiary is reviewing that subsidiary's forecasts for the next few months, and she has identified that there will be a short term deficit of cash starting on 31 October. She is therefore planning to borrow \$10 million for a six-month period starting on 31 October.

Bank base rates in South Asia are currently 4%, and the company's credit rating is such that it pays a credit spread of 20 basis points on its borrowings.

There has been a lot of volatility in recent months and the finance manager thinks there is a possibility that bank base rates could increase by 1% or decrease by 1% before 31 October. She is therefore keen to hedge against this risk using traded derivative products.

The following market information is available:

Futures contracts – prices for \$1 million, 3-month contracts

June	96.05
Sept	95.68
Dec	95.16

Options over December futures contracts – \$1 million 3-month contracts, premia are quoted in %

	CALLS	PUTS
Strike price	Dec	Dec
95.00	N/A	0.25
95.50	0.22	N/A

It can be assumed that settlement for the futures and options contracts is at the end of the month, and that basis diminishes to zero at contract maturity at a constant rate.

The South Asian subsidiary finance manager is concerned that the options over futures contracts are very expensive to set up, and she feels that the head office finance team would not authorise the expenditure on the premium payable. Therefore, she wants to explore the possibility of a futures hedge and a collar hedge, but not a traditional options hedge.

(a) Calculate the results of the hedging strategies that are being considered for the \$10 million borrowing, if the bank base rate increases by 1% or decreases by 1%. Show all relevant calculations and comment on your results.

(15 marks)

(b) Discuss the advantages and disadvantages of Lamontagne Co setting up a centralised treasury function.

(5 marks)

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, and commercial acumen in your answer.

(5 marks)